Caleb Chen

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Objective

Electrical Engineering student seeking an ECE internship to apply hands-on experience in hardware validation, test automation, embedded systems, and electrical prototyping. Experienced in troubleshooting electrical systems, developing Python-based automated tests, using laboratory instrumentation, and programming microcontrollers. Brings strong hardware, software, and problem-solving skills with a focus on developing reliable and thoroughly tested engineering solutions.

Education

Transfer to Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Electrical Engineering | Transfer with 84 Credit Hours Expected Graduation, May 2028

Auburn University | Auburn, AL *August 2025 – May 2026*
Bachelor of Science in Electrical Engineering | GPA 4.0
Coursework: Circuit Analysis, Data Structures and Algorithms

Skills

Programming & Automation: Python (OOP), Linux(Ubuntu), SPI, UART, SCPI, Regex, automated testing, firmware flashing
Hardware: Hardware validation, failure analysis, PCB assembly, soldering, wiring, microcontrollers, sensors, gateways, 0–10V dimming
Instrumentation: Oscilloscopes, digital multimeters, Keithley, BK Precision, California Instruments, power meters, environmental chambers
Design & Workflow: Fusion 360, KiCad, Lush, MATLAB, 3D printing, Containers, Excel, Jira, Confluence, Agile, AI-assisted workflows

Experience

Synapse Wireless | Huntsville, AL *Fall 2024, Summer 2025, Summer 2026*
Electrical Engineer Intern

- Supported product validation by testing 5000+ ESD sensitive radio modules, gateways, PiNet systems, and other embedded devices, including SD card programming and MAC address reconfiguration as needed.
- Conducted electrical validation tests to verify emergency power transition performance, measuring ~220ms average power rail hold-up time and confirming 100% system operation during emergency AC power loss and backup inverter switchover.
- Utilized Fusion 360 to create 3D-printed testing stations with detailed technical documentation to audit and triage firmware, scripts, sensor inputs, dimming, and link quality of 200+ lighting control units, leveraging Linux/Ubuntu-based tools and Python to control power supplies, multimeters, and other lab instrumentation.
- Assisted qualification of 4 new products' hardware (DVT) through electrical, thermal, and mechanical testing.
- Enhanced team efficiency by standardizing FA data collection, reducing bottlenecks on the Failure Analysis rack and clearing roughly 50 backlogged RMAs, directly contributing to faster customer return cycles.
- Converted customer requirements into a scalable Excel tool that structured site data for lighting setup, accelerating these processes by 80% and reducing risks of error.

Projects

Residential Solar Energy System *Summer 2026*
Independent Renewable Energy Project

- Designed and independently installed a residential solar energy system with 61.4 kWh of $LiFePO_4$ battery storage.
- Installed the complete 48 V DC power distribution, high-current AC/DC interconnects, and dual-MPPT solar charge-controller integration.
- Conducted AC wiring topology for a 10 kW split-phase inverter

MIDI-Controlled Addressable LED System *Summer 2026*
Independent Embedded Systems Project

- Designed an embedded lighting system using an ESP32-S3 to receive and process real-time MIDI signals from a digital piano.
- Developed firmware to decodes and translates MIDI note, velocity data, and musical events into dynamic customizable addressable LED patterns.

Self-Hosted Photo Backup Server (Docker) *Fall 2026*
Independent Systems & Networking Project

- Deployed and maintained a headless Linux-hosted photo backup server (Immich) that automatically syncs 2,000+ photos from my phone to local storage, with no third-party cloud dependency.
- Containerized the application, database, and machine-learning services with Docker Compose, using private virtual networks and persistent volumes so data survives updates and container failures.
- Diagnosed networking and service issues (ports, IP addressing, startup order, permissions) using Linux CLI tools and container logs, and secured remote access with Tailscale VPN while limiting exposed ports.